College Physics 2

Electric Potential and Potential Difference

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College Physics 2

Work and Electric Potential Energy

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OBJECTIVES

- Define electric potential and electric potential energy
- Calculate the work done in moving a charge in a region of constant electric field.

ELECTRIC POTENTIAL ENERGY

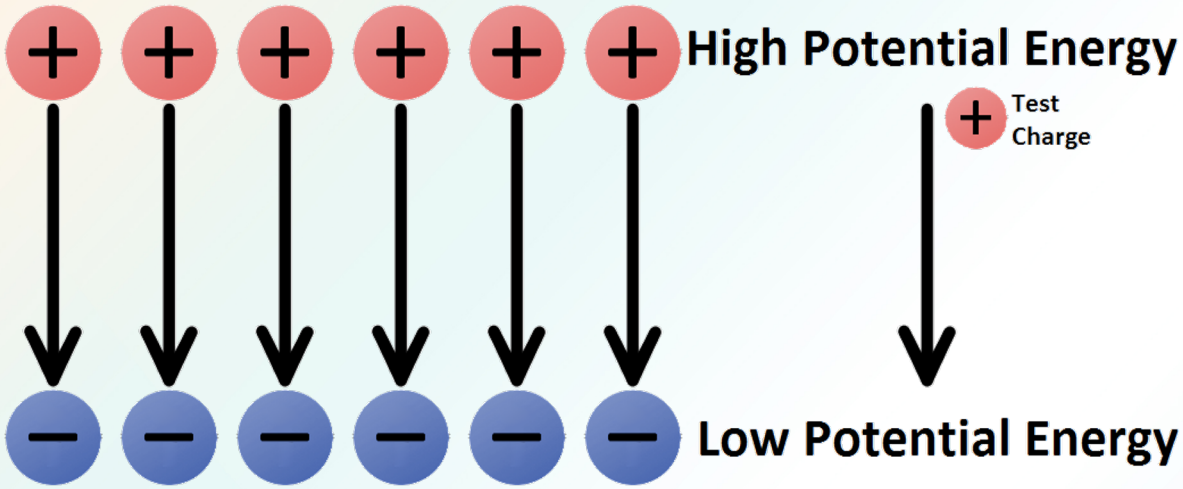


Figure 1. Electric potential energy. From.*https://cdn.sparkfun.com/r/600-600/assets/2/3/5/1/e/51a4f1a7ce395f4125000000.png*

Electric potential energy is the energy that charged particles such as electrons and protons have because of its own electric charge and its relative position to other charged particles.

Recall

Work is a scalar quantity equal to the product of the displacement x and the component of the force F_x in the direction of the displacement.

$$Work = F_x x$$

$$Work = (F \cos \theta) x$$

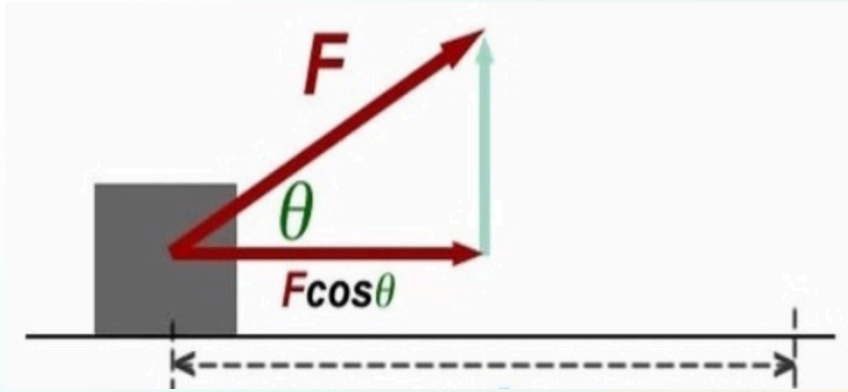


Figure 2. Work done at a certain angle. From. *https://qph.fs.quoracdn.net/main-qimg-5dd99f3fd1d9aff67e96375dc46f61ff*

Recall

Potential Energy: Ability to do work by virtue of position or condition.

$$U = mgh$$

Kinetic Energy: Ability to do work by virtue of motion. (Mass with velocity)

$$K = \frac{1}{2}mv^2$$

The Work-Energy Theorem: The work done by a resultant force is equal to the change in kinetic energy that it produces.

$$\text{Work} = \tfrac{1}{2}\,mv_f^2 - \tfrac{1}{2}\,mv_o^2$$

ELECTRIC POTENTIAL ENERGY

Potential Energy and Work

$$PE = mgy$$

$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$$

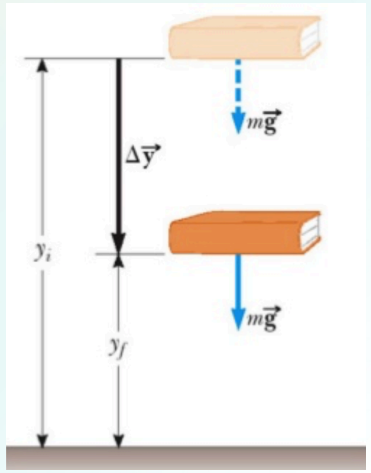


Figure 3. Work and potential energy. From. *https://kaiserscience.files.wordpress.com/2015/09/work-energy-and-power-30-638.jpg*

$$\Delta PE_{a\to b} = mgy_b - mgy_a = mg\Delta y = F_G\Delta y$$

$$W_{a\to b} = F_A(y_b - y_a) = F_G\Delta y$$

The amount of work done is equal to the change in potential energy.

ELECTRIC POTENTIAL ENERGY

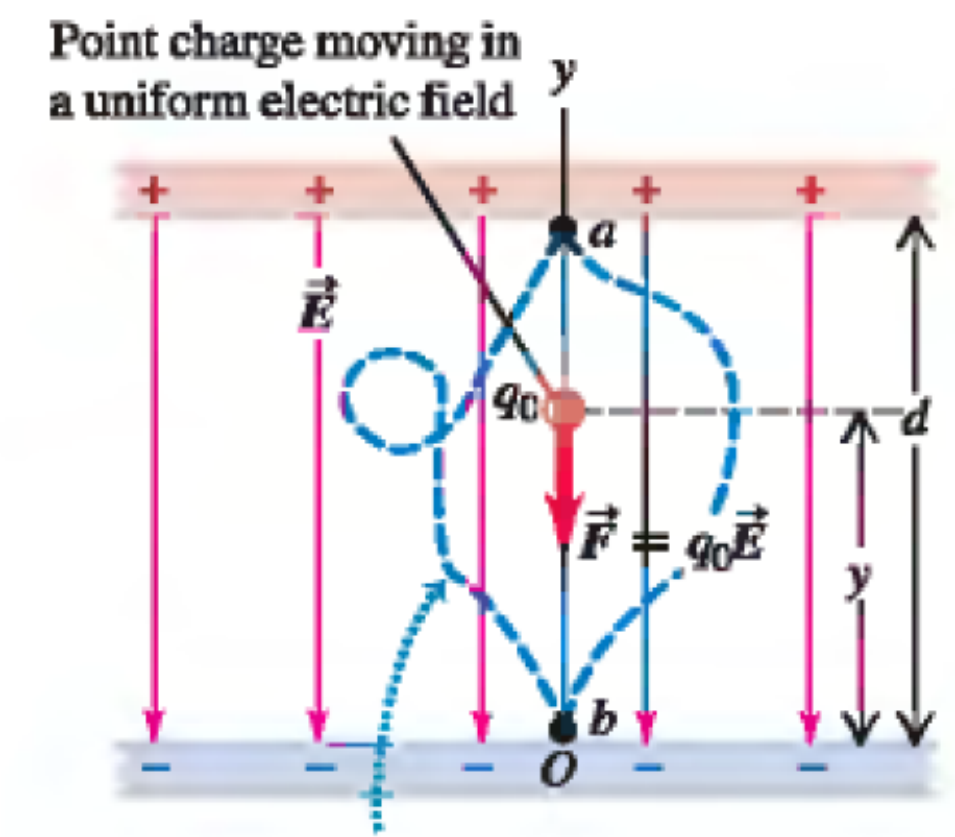


Figure 4. Work done by the electric force. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 755) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

- The field exerts a downward force on the positive test charge.
- The charge moves from point **a** to **b.**
- Work done by the field is positive.

$$W_{a\to b} = q_0 E(y_b - y_a) = q_0 Ed$$

ELECTRIC POTENTIAL ENERGY AND WORK (done by e-field)

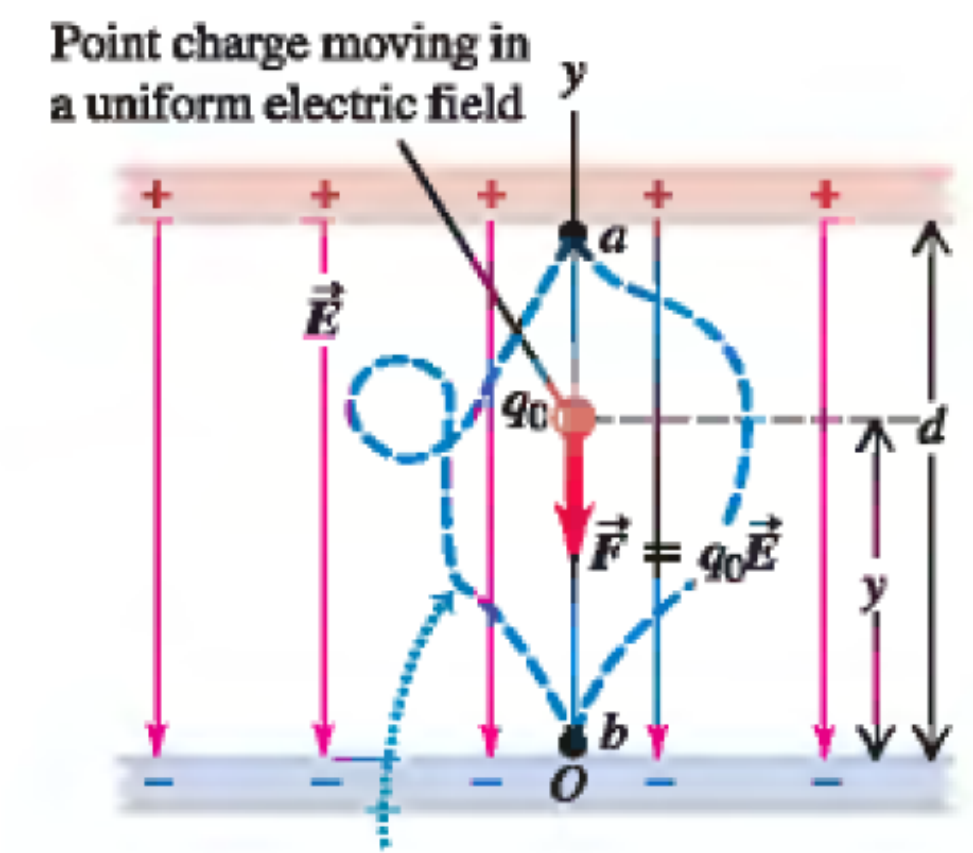


Figure 4. Work done by the electric force. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 755) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

- Since work done by the field as the charge moves from point **a** to **b** is equal to the change in potential energy from **a** to **b**,

$$\Delta U = q_0 E y_b - q_0 E y_a$$
$$= q_0 E(-d) = -W_{a\to b}$$

$$W_{a\to b} = -\Delta U$$

ELECTRIC POTENTIAL ENERGY AND WORK (done by e-field)

$$W_{a\to b} = -\Delta U$$

- When work done by the field is **positive**, potential energy **decreases**.
- When work done by the field is **negative**, potential energy **increases**.

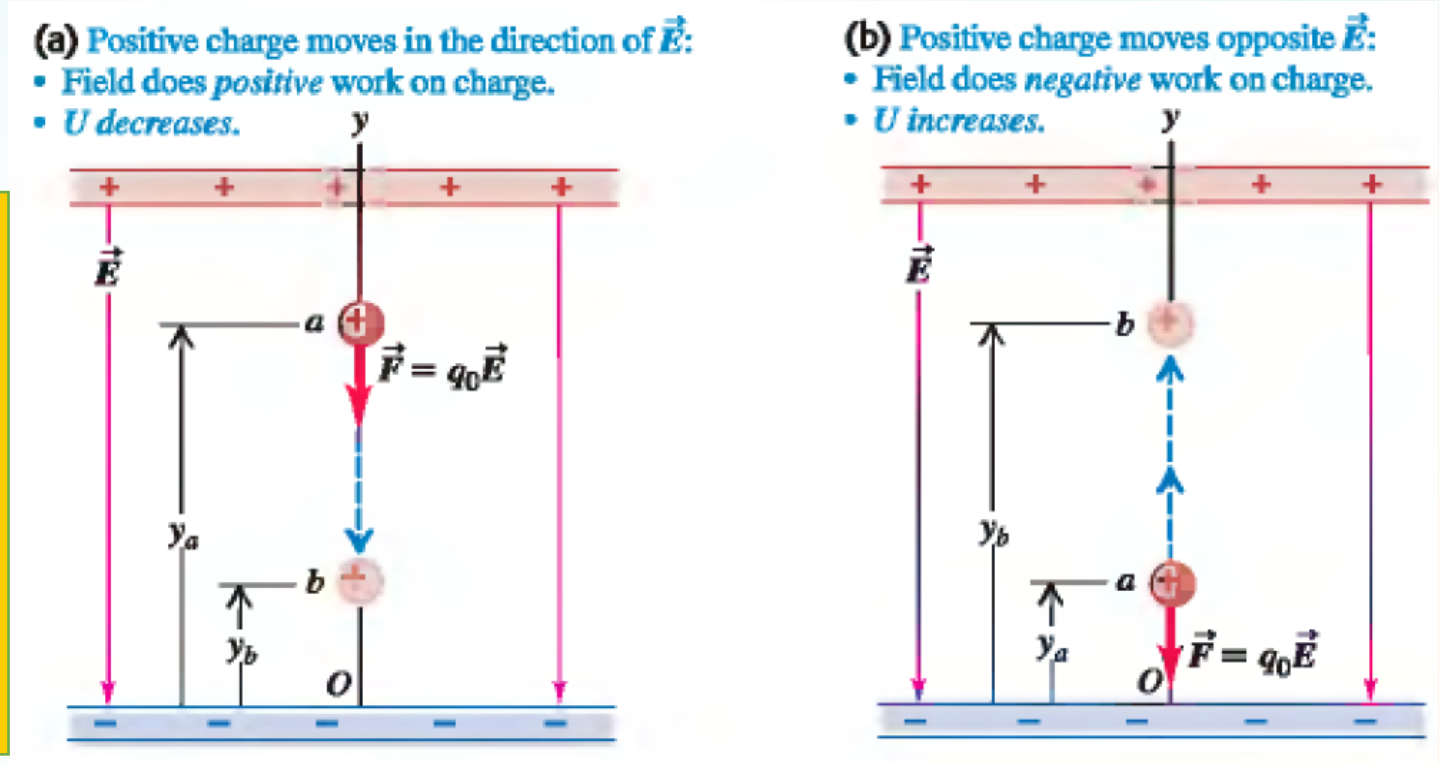


Figure 5. Motion of positive charge. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 756) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

ELECTRIC POTENTIAL ENERGY AND WORK (done by e-field)

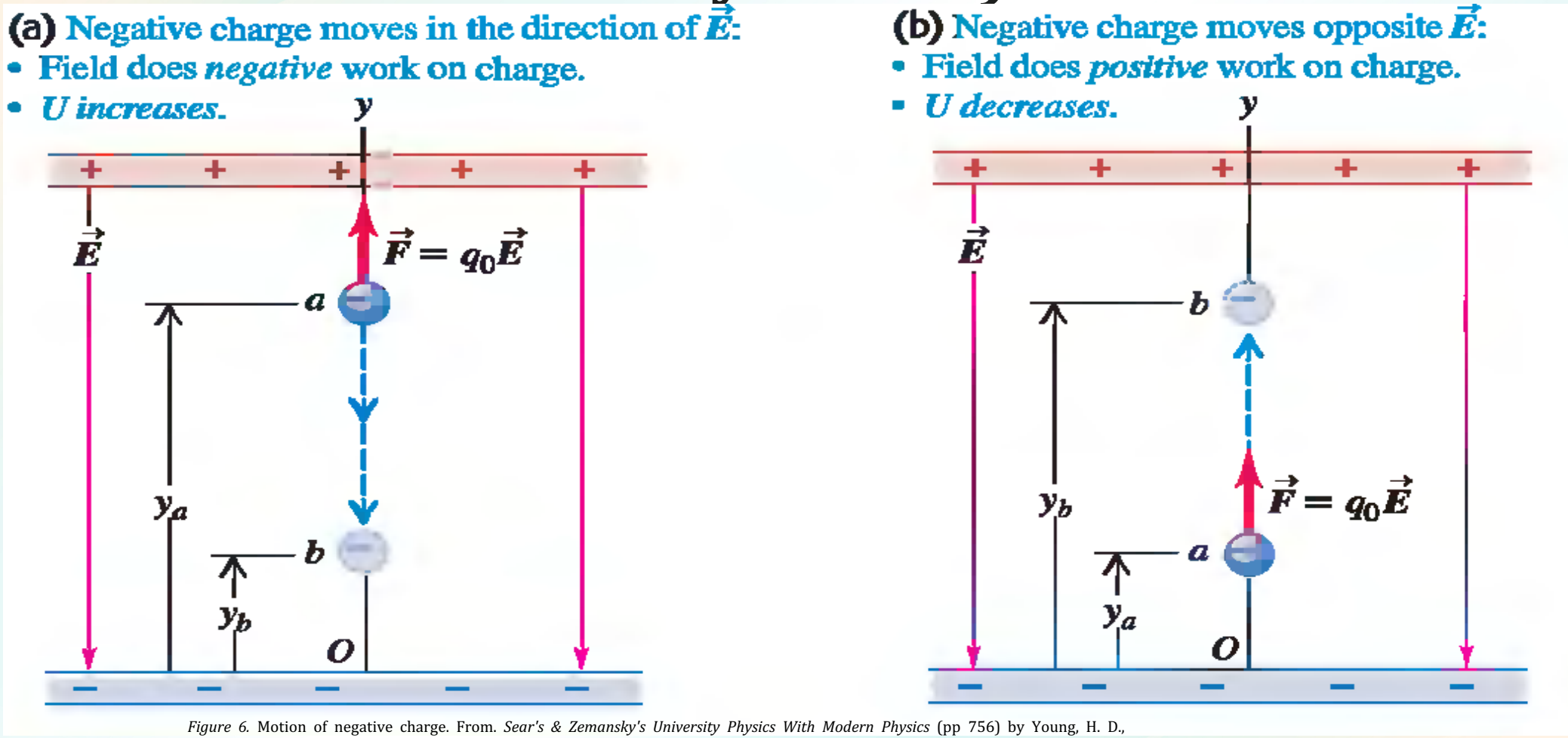
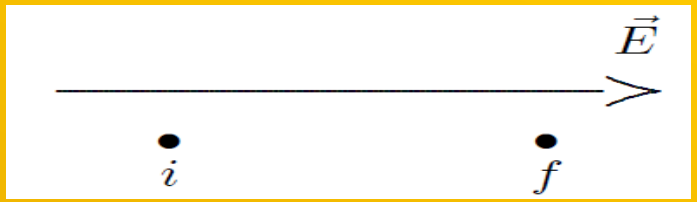


Figure 6. Motion of negative charge. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 756) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

ELECTRIC POTENTIAL ENERGY AND WORK (done by e-field)

Conceptual Question

A proton moves from point i to point f, in the direction of a uniform electric field. During this displacement:

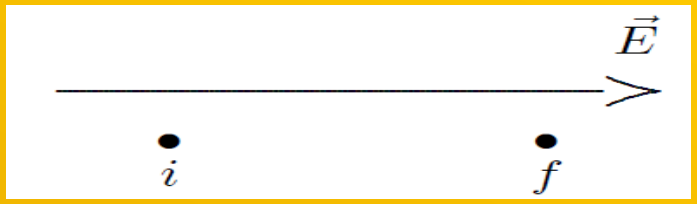


a. the work done by the field is positive and the potential energy of the electron-field system increases
b. the work done by the field is negative and the potential energy of the electron-field system increases
c. the work done by the field is positive and the potential energy of the electron-field system decreases
d. the work done by the field is negative and the potential energy of the electron-field system decreases

ELECTRIC POTENTIAL ENERGY AND WORK (done by e-field)

Conceptual Question

A proton moves from point i to point f, in the direction of a uniform electric field. During this displacement:



a. the work done by the field is positive and the potential energy of the electron-field system increases
b. the work done by the field is negative and the potential energy of the electron-field system increases
c. the work done by the field is positive and the potential energy of the electron-field system decreases
d. the work done by the field is negative and the potential energy of the electron-field system decreases

Potential Energy of 2 point charges

Force on the test charge due to the charge q

$$\vec{F} = k\frac{qq_0}{r^2}\hat{r}$$

The work done on by in moving from a to b

$$W_{a\to b} = \int_a^b k\frac{qq_0}{r^2}\hat{r}\cdot d\vec{r} = kqq_0\int_a^b \frac{dr}{r^2}$$

$$W_{a\to b} = kqq_0\left(-\frac{1}{r}\Big|_a^b\right) = kqq_0\left(-\frac{1}{b}+\frac{1}{a}\right)$$

$$W_{a\to b} = kqq_0\left(\frac{1}{a}-\frac{1}{b}\right)$$

Depends only on the endpoints

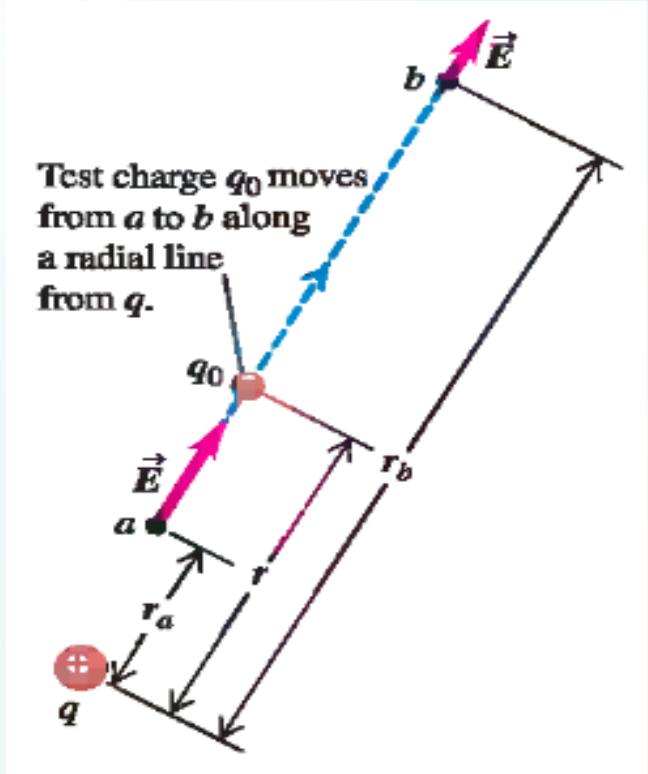


Figure 7. Electric potential energy of two point charges. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 756) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Note on Potential Energy

- Potential energy is always defined **relative to some reference point** where U=0

$$U = k\frac{qq_0}{r} \rightarrow 0$$ If q and are infinitely apart

- U represents the **work done <u>by the electric field</u> (of q)** in moving q_0 from r to ∞

Work and Potential Energy

- Work that is done by the electric force when the particle moves from a - b

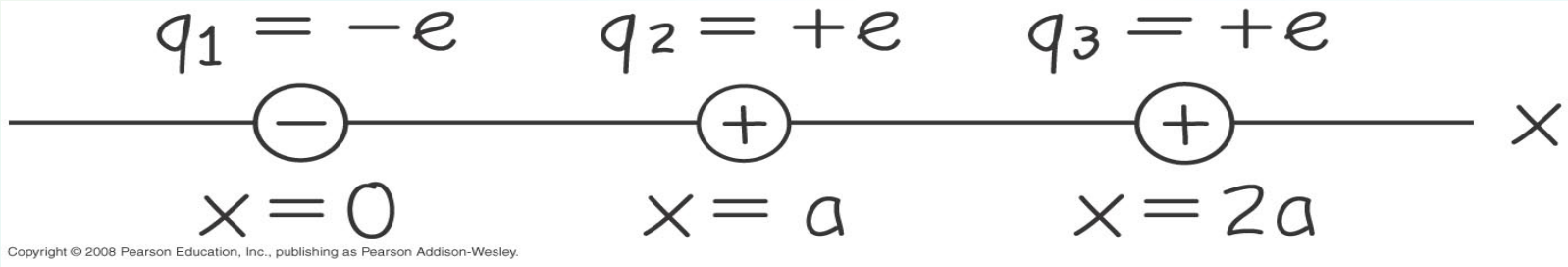
$$W_{electric} = U_a - U_b = -\Delta U$$

- Work that must be done by an external force to move the particle from a – b against the electric force

$$W_{external} = -W_{electric} = U_b - U_a = \Delta U$$

Sample Problems

Find the work that must be done by an external force to bring a third point charge q_3=+e from infinity to x=2a.



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$$W_{external} = U_{final} - U_{initial}$$

$$W_{q_3} = U_{x=2a} - U_{x=\infty} = ke\left(-\frac{e}{2a} + \frac{e}{a}\right) - 0$$

$$W_{q_3} = +\frac{ke^2}{2a}$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

REFERENCES

https://cdn.sparkfun.com/r/600-600/assets/2/3/5/1/e/51a4f1a7ce395f4125000000.png

https://qph.fs.quoracdn.net/main-qimg-7cd2ff936c83c099652dd85e2c2d77bb

https://kaiserscience.files.wordpress.com/2015/09/work-energy-and-power-30-638.jpg

College Physics 2

Electric Potential and Electric Potential Difference

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OBJECTIVES

- Define electric potential and electric potential difference
- Calculate electric potential of a collection of charges
- Use equipotential surfaces to visualize how the electric potential varies in space

ELECTRIC POTENTIAL ENERGY

The electric potential energy of a test charge $\mathbf{q_0}$ in a field produced by charge $\mathbf{q}$ is given by

$$U = \frac{kq_0q}{r}$$

where:

U is the electric potential energy in **joules (J)**

$k = \frac{1}{4\pi\epsilon_0} = 9.0 \text{ x } 10^9 \text{ Nm}^2/\text{C}^2$

$\mathbf{r}$ is the distance from $\mathbf{q_0}$ to $\mathbf{q}$ (from test charge to source) in $\mathbf{m}$

For a system of point charges, the electric potential energy is given by:

$$U = k\left(\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_1q_3}{r_{13}} + \cdots\right)$$

Electric Potential Energy for **several** point charges

- Total electric field is the vector sum of individual charges
- Work done is just the sum of the contribution of individual charges

$$U_a = kq_o\left(\frac{q_1}{r_1}+\frac{q_2}{r_2}+\frac{q_3}{r_3}+\ldots\right)=kq_o\sum_i\frac{q_i}{r_i}$$

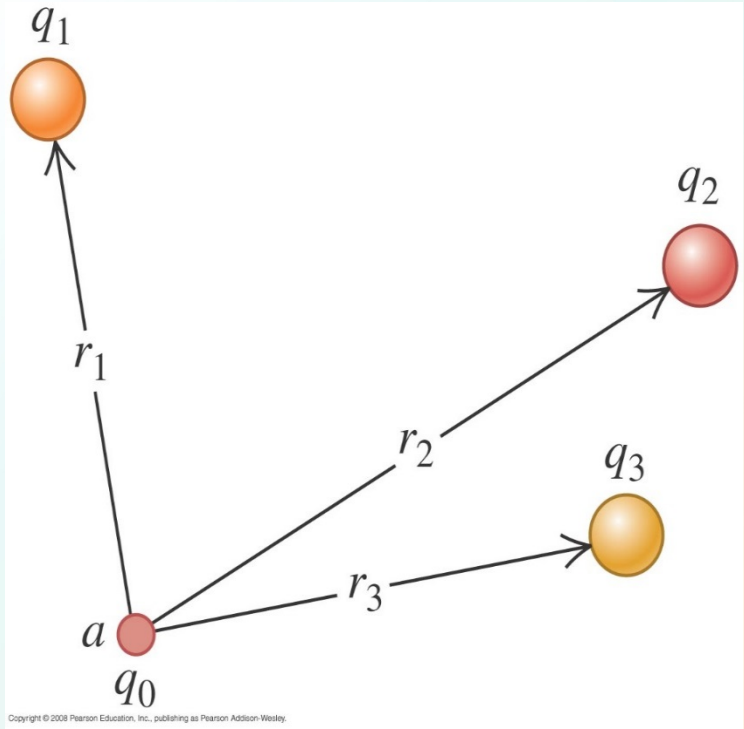


Figure 8. Potential Energy for Several Point Charges.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 759) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

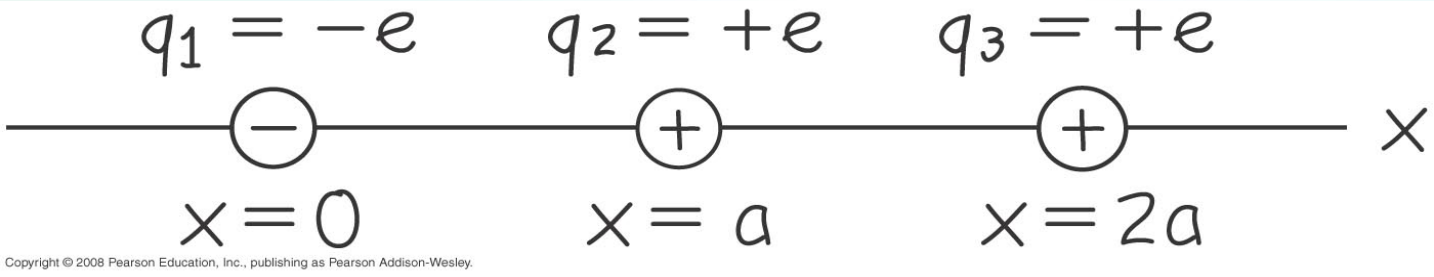
Potential Energy in assembling charges q_i

$$U = k\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} + \ldots\right) = k\sum_{i<j}\frac{q_i q_i}{r_{ij}}$$

- Sum extends over all pairs
- A charge cannot interact with itself (i=j) not included
- Avoid over-counting (i<j)

Sample Problems

1. Find the **total potential energy** of the system of three charges.



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$$U = k\sum_{i<j}\frac{q_i q_j}{r_{ij}} = k\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$= k\left(\frac{(-e)(e)}{a} + \frac{(-e)(e)}{2a} + \frac{(e)(e)}{a}\right)$$

$$U = -\frac{ke^2}{2a}$$

Sample Problems

2. A point charge $q_1 = +4.00\,nC$ is placed at the origin, a second point charge $q_2 = -3.00\,nC$ is placed on the x-axis at $x = 20.0cm$ and a third charge $q_3 = +2.00\,nC$ is placed at $x = 10.0\,cm$. What is the potential energy of the system of three charges?

Solution:

$$U_T = U_{12} + U_{23} + U_{13}$$

$$U_T = K_e\frac{q_1q_2}{r_{12}^2} + K_e\frac{q_2q_3}{r_{23}^2} + K_e\frac{q_1q_3}{r_{13}^2}$$

$$U_T = 9\times10^9\left(\frac{4\times10^{-9}\cdot-3\times10^{-9}}{20\times10^{-2}} + \frac{-3\times10^{-9}\cdot2\times10^{-9}}{10\times10^{-2}} + \frac{4\times10^{-9}\cdot2\times10^{-9}}{10\times10^{-2}}\right) = \boxed{-3.6\times10^{-7}\text{J}}$$

ELECTRIC POTENTIAL

$$V = \frac{U}{q} = \frac{kq}{r}$$

Where

V = electric potential (V)
U =Potential energy (J)
q = charge (C)
r = distance of the charge (m)
k = proportionality constant
k = $9x10^9$ Nm^2/C^2

Electrical Potential, V, of a point charge is the ratio of the total electric potential energy and the amount of the charge.

ELECTRIC POTENTIAL DIFFERENCE

Unit is volt.

Volt is defined as the change in electrical potential difference that occurs when 1-coulomb charge experiences a change in electrical potential energy of one joule.

$$1\,Volt = \frac{1 Joule}{Coulomb}$$

Electrical potential difference or voltage change (or simply Voltage) is defined as the change in electrical energy a particle undergoes when moving between different points in space.

ELECTRIC POTENTIAL DIFFERENCE

$$\frac{W_{a\to b}}{q_0} = -\frac{\Delta U}{q_0} = -\left(\frac{U_b}{q_0} - \frac{U_a}{q_0}\right) = -(V_b - V_a) = V_a - V_b = V_{ab}$$

This means that the work done by the electric force to move a charge from point **a** to **b** is equal to the **potential difference *Vab*.**

ELECTRIC POTENTIAL DIFFERENCE

HOW DO BATTERIES WORK?

- The positive terminal of the battery represents the point with high potential while the negative terminal represents the point with low potential.
- The voltage of the battery is the potential difference between the terminals. This potential difference is needed for the charges to flow.

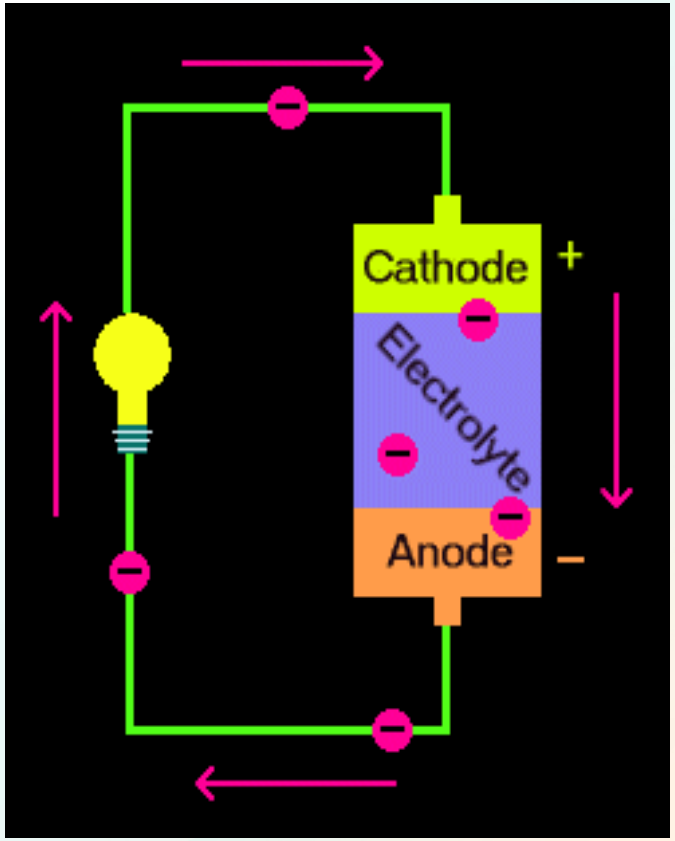


Figure 9. How do batteries work. From *http://www.qrg.northwestern.edu/projects/vss/docs/media/Power/battery.gif*

ELECTRIC POTENTIAL DIFFERENCE

Electrical sources such as batteries and generators supply a constant potential difference.

TABLE 23–1 Some Typical Potential Differences (Voltages)

Source	Voltage (approx.)
Thundercloud to ground	10^8 V
High-voltage power line	10^5–10^6 V
Power supply for TV tube	10^4 V
Automobile ignition	10^4 V
Household outlet	10^2 V
Automobile battery	12 V
Flashlight battery	1.5 V
Resting potential across nerve membrane	10^{-1} V
Potential changes on skin (EKG and EEG)	10^{-4} V

ELECTRIC POTENTIAL

The potential due to a single point charge **q** is given by

$$V = \frac{kq}{r}$$

where:

V is electric potential in **V**

$k = \frac{1}{4\pi\epsilon_0} = 9.0 \text{ x } 10^9 \text{ Nm}^2/\text{C}^2$

r is the distance from **q** to the point where the potential is evaluated

For a collection of point charges, the potential is given by

$$V = k\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \cdots\right)$$

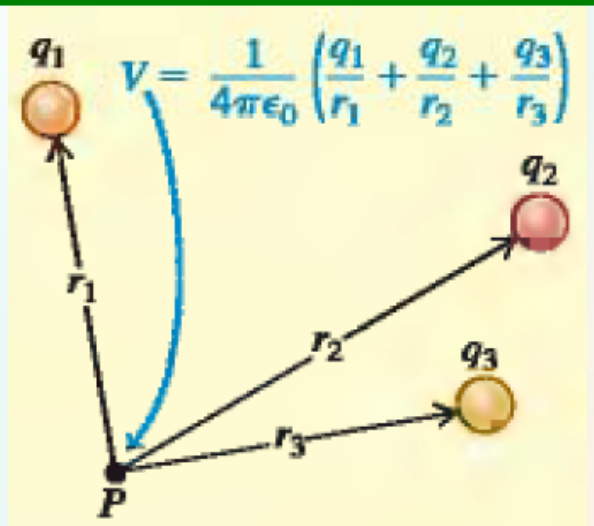


Figure 10. Potential Energy for Several Point Charges.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 777) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Sample Problem

3. A point charge has a charge of $2.50\times10^{-11}C$. What is the electric potential at a point 1.75 m away?

Solution:

$$V = \frac{U}{q} = \frac{kq}{r}$$

$$V = \frac{(9x\ 10^{9}\ )(\ 2.5\ x\ 10^{\ -11})}{1.75\ m} = \mathbf{0.129\ Volt}$$

Sample Problem

4. Determine the potential at a point 0.50m from
   A. a +20 μC point charge.
   B. a -20 μC point charge.

Solution:

A:

$$V = \frac{kq}{r}$$

$$V = \left(\frac{9x10^{9}Nm^{2}/C^{2})(20x10^{-6}C)}{0.50\ m}\right)$$

$$\boldsymbol{V = 3.6\ x10^{5}\ V}$$

B:

$$V = \frac{kq}{r}$$

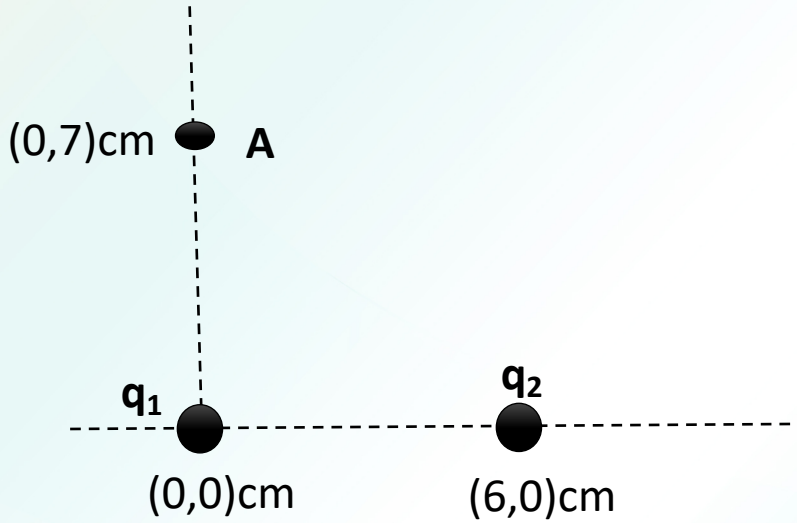
$$V = \left(\frac{9x10^{9}Nm^{2}/C^{2})(-20x10^{-6}C)}{0.50\ m}\right)$$

$$\boldsymbol{V = -3.6\ x10^{5}\ V}$$

Sample Problem

5. In the Cartesian coordinate system, $q_1 = 2x10^{-9}C$ is at the origin and $q_2 = 6x10^{-9}C$ is placed at (6,0) cm.. Find the net potential at points A (0,7)cm.

Solution:



$$V_1 = \frac{kq_1}{r} = \left(\frac{9x10^9 Nm^2/C^2)(2x10^{-9}C)}{0.07\ m}\right) = 257.14\ V$$

$$V_2 = \frac{kq_2}{r} = \left(\frac{9x10^9 Nm^2/C^2)(6x10^{-9}C)}{0.092\ m}\right) = 586.96\ V$$

$$V_{net}at\ A = V_1 + V_2 = (257.14 + 586.96)V$$

$$\boldsymbol{V_{net}at\ A = 844.10\ V}$$

EQUIPOTENTIAL SURFACE

The potential difference between 2 points is equal to zero.

No work is required to move a charge from one point to another on an equipotential surface.

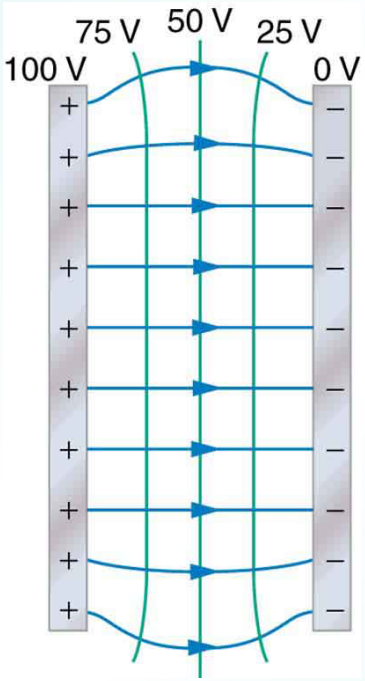


Figure 11. Equipotential Surface. From. *https://s3-us-west-2.amazonaws.com/courses-images/wp-content/uploads/sites/1989/2017/06/13225944/figure-20-04-04a.jpeg*

> An **equipotential surface** is one on which all points are at the same potential.

EQUIPOTENTIAL SURFACE

An **equipotential surface** is a three-dimensional surface on which the *electric potential V* is the same at every point.

If a test charge q_0 is moved from point to point on such a surface, the **electric potential energy remains constant**.

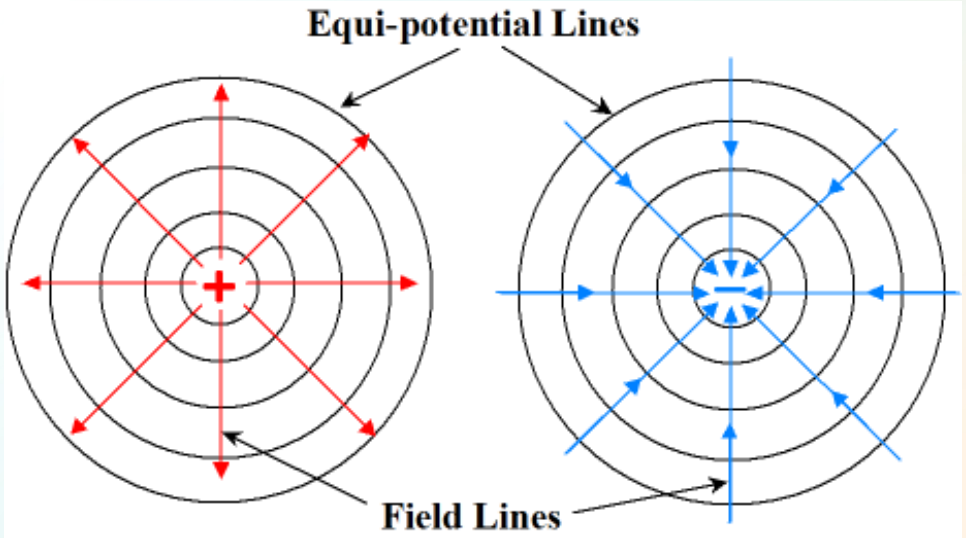


Figure 12. Equiotential surface. From. *https://kaiserscience.files.wordpress.com/2016/01/equipotential-lines.gif*

EQUIPOTENTIAL SURFACE

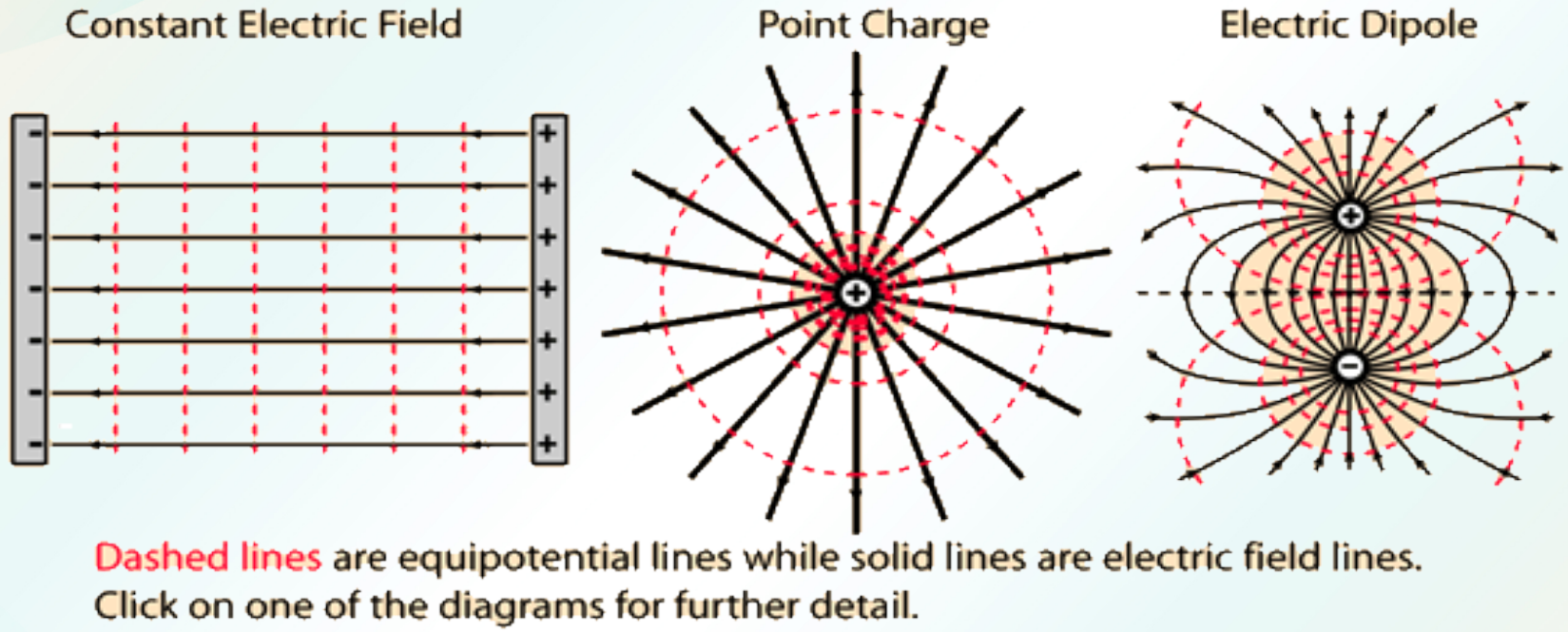


Figure 13. Equipotential Surfaces. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp755) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

An equipotential surface must be perpendicular to the electric field at any point.

Sample Problem

For a single point charge with $q = 4.0 \times 10^{-9}$ C, sketch the equipotential surfaces (or lines in a plane containing the charge) corresponding to $V_1 = 10$ V, $V_2 = 20$ V, and $V_3 = 30$ V.

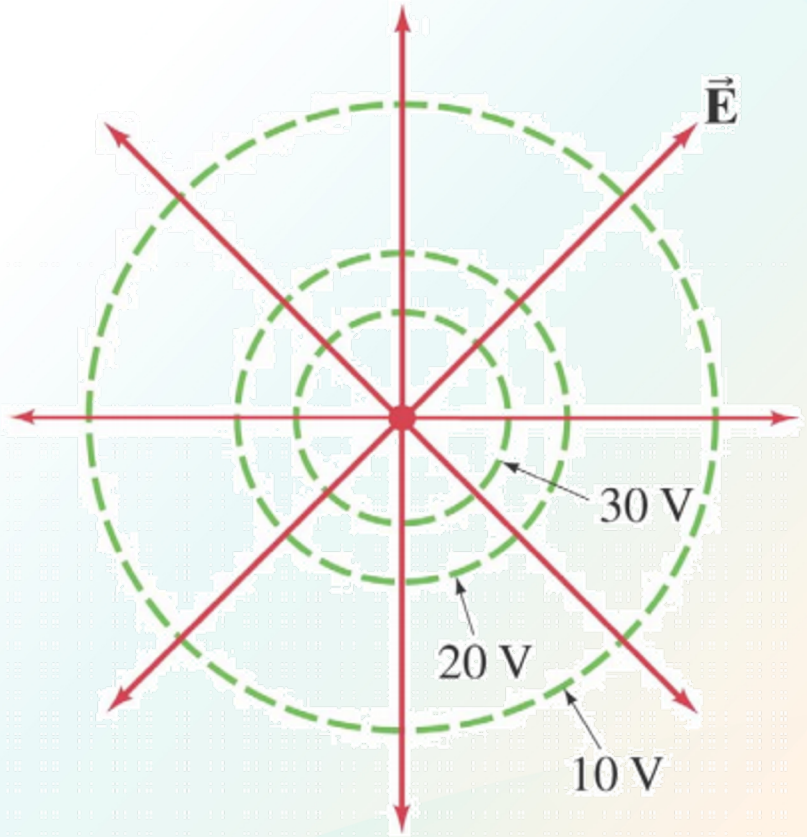
Sample Problem

Solution:

$$V = \frac{kq}{r}$$

So,

$$r = \frac{kq}{V}$$



For: $V_1 = 10$ V, $r = 3.6$ m
$V_2 = 20$ V, $r = 1.8$ m
$V_3 = 30$ V, $r = 1.2$ m

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

http://www.qrg.northwestern.edu/projects/vss/docs/media/Power/battery.gif

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